CODEINE

Not for children <12 yr

Controlled drug

CHEMICAL NAME (5α,6α)-7,8-didehydro-4,5-epoxy-3-methoxy-17-methylmorphinan-6-ol (as phosphate)

CLASS Weak opioid agonist (prodrug of morphine); antitussive

TRADE NAMES Codeine phosphate · Solpadeine (with paracetamol) · Co-codamol · Tylenol with Codeine

THERAPEUTIC USES / INDICATIONS

- Mild to moderate pain (often combined with paracetamol).
- Dry, unproductive cough.
- Acute diarrhoea (short-term).

ADMINISTRATION

- **PO** tablets, syrup; can be given with food to reduce nausea.
- **IM** injection (rarely used).

DOSAGES

- **Adult pain:** 30–60 mg every 4–6 h; max 240 mg/day.
- **Cough:** 15–30 mg 3–4 times daily.
- **Children 12–18 yr:** 30–60 mg every 6 h (max 240 mg/day).
- Limit to 3 days for acute pain in adolescents.

PHARMACOKINETICS

- **Onset:** 30–60 min. **Peak:** 1–1.5 h. **Duration:** 4–6 h.
- Prodrug: ~10% converted to morphine by CYP2D6 in the liver.
- Excreted in urine. **t½:** about 3 h.

PHARMACODYNAMICS

- Acts mainly through morphine at **μ-opioid receptors**.
- Suppresses cough centre in the medulla.
- Reduces GI motility (antidiarrhoeal).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Constipation, nausea, vomiting.
- Drowsiness, dizziness.
- Dry mouth, itching.

ADVERSE EFFECTS (SERIOUS)

- Respiratory depression — especially in CYP2D6 ultra-rapid metabolisers (deaths in children).
- Dependence with prolonged use.
- Biliary spasm; urinary retention.
- Overdose: coma, miosis, respiratory depression.

CONTRAINDICATIONS & CAUTIONS

- Children <12 yr; under 18 after tonsillectomy/adenoidectomy.
- Known CYP2D6 ultra-rapid metabolisers; breastfeeding.
- Acute respiratory depression; acute asthma.
- Paralytic ileus; raised intracranial pressure.

KEY INTERACTIONS

- **CNS depressants, alcohol:** additive sedation.
- **CYP2D6 inhibitors** (fluoxetine, paroxetine): ↓ analgesic effect.
- **Anticholinergics:** ↑ constipation, urinary retention.

NURSING MANAGEMENT

- Check age and indication before giving.
- Monitor pain, RR, sedation and bowel function.
- Give laxative with regular use.
- Check total paracetamol dose in combination products.
- Assess cough frequency and character when used as antitussive.

PATIENT EDUCATION

- May cause drowsiness; avoid driving and alcohol.
- Increase fluids and fibre to prevent constipation.
- Do not use in children under 12.
- Not for long-term use without review.

KEY POINTS

- Prodrug of morphine via CYP2D6.
- Contraindicated <12 yr and in breastfeeding.
- Constipation is very common.
- Max 240 mg/day.